W-G-441A
November 19, 1987
SUPERSEDING
W-G-441
September 27, 1951

FEDERAL SPECIFICATION

GLUEPOTS, ELECTRIC (PORTABLE, DRY AND WET)

This specification is approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers a portable electrical gluepot in which animal glue is heated to a preset temperature and automatically maintained by means of a thermostat and electrical coils.

1.2 Classification.

1.2.1 Types and sizes. Pots shall of the following types and sizes with automatic temperature control, as specified.

Type I - Dry type
Type II - Wet type (water jacket)

Sizes.

Size 1 - One quart capacity
Size 2 - Two quart capacity
Size 3 - Four quart capacity

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

MMM-A-100 - Adhesives, Animal Glue
PPP-P-40 - Packaging and Packing of Hand Tools

(Activities outside the Federal Government may obtain copies of Federal specifications, standards, and commercial item descriptions, as outlined under General Information in the Index of Federal Specifications, Standards and Commercial Item Descriptions. The index, which includes cumulative bimonthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification, other Federal specifications, standards, and commercial item description required by activities outside the Federal Government for bidding purposes are available without charge from General Services Administration Business Service Centers in Boston; New York; Philadelphia; Washington, DC; Atlanta; Chicago; Kansas City, MO; Fort Worth; Houston; Denver; Los Angeles; San Francisco; and Seattle, WA.

(Federal Government activities may obtain copies of Federal specifications, standards, commercial item descriptions and the Index of Federal Specifications, from established distribution points in their Agencies.)

Military Standard:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of military specifications and standards required by contractors in connection with specific procurement functions should be obtained from procuring activity or as directed by the contracting officer.)

FSC 5120

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American Society For Testing and Materials (ASTM) Standards:

ASTM B 344 - Specification for Drawn or Rolled Nickel-Chromium And Nickel-Chromium Iron Alloys for Electrical Heating Elements.

(Application for copies should be addressed to the American Society For Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

Underwriters' Laboratories, Inc. Standards:

No. 498 - Attachment Plugs and Receptacles.
No. 499 - For Electric Heating Appliances.

(Copies of Underwriters' Laboratories Standards may be obtained from Underwriters' Laboratories Inc., 1285 Walt Whitman Road, Melville, Long Island, NY 11747.)

3. REQUIREMENTS

3.1 Fire and casualty hazards. Electric gluepots delivered under this specification shall comply with the applicable requirements of Underwriters' Laboratories Standard No. 499. The bidders shall submit proof to the purchasing agency that the electric gluepots he proposes to furnish under this specification conform to the standards of the Underwriters' Laboratories Inc. regarding fire and casualty hazards. The label or listing of the Underwriters' Laboratories will be accepted as conforming to this requirement. The bidder may submit independent proof satisfactory to the purchasing agency that the electric gluepots conform to the published standards including method of test of the Underwriter Laboratories.

3.1.1 Compliance with the above preliminary requirements, regarding fire and casualty hazards, does not absolve the bidder from complete compliance with the other requirements of this specification in order to secure the acceptance of his electric gluepots.

3.2 General description. Each gluepot shall consist of a readily removable inner pot and portable electric heater housing with flexible supply cable.

3.3 Voltage rating and frequency. Pots shall operate on either 115 or 230 volts and on either direct or alternating 60-cycle current, as specified in the invitation for bids.

3.4 Heater housing.

3.4.1 General description. The heater housing shall be compact and substantial in construction. Outer surfaces, if of iron or steel, shall be galvanized or japanned.

3.4.2 Handle. Each pot shall be provided with a bail-type handle.

3.4.3 Type I, dry type. The housing shall be of double-wall construction. The inner wall shall be of a seamless durable metal and shall be plated to resist corrosion. Insulation shall be used between the outer and inner walls.

3.4.4 Type II, wet type. The housing may be of single- or double-wall construction. The surface in contact with the water must be impervious to water and must not be adversely affected by potable water. Insulation shall be used in the walls of double construction.

3.5 Inner pot. Unless otherwise specified (see 6.1), the inner pot shall be of seamless construction and shall be of copper, copper-nickel alloy, aluminum, aluminum alloy or corrosion-resisting steel, and shall have a smooth polished interior. It shall be provided with an attached bail handle made of such material and design as to provide adequate personnel protection from accidental burns. Brush wiper, and top flange shall extend beyond the outer edge of the outer pot. All material shall be of such quality and thickness as to be rigid and durable.

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Unless the gluepot is of such design as to prevent glue from seeping between the inner pot and inner side of heater, a reinforcement or other means shall be fitted on the inner pot in order to permit its forcible removal from the housing in case the pot sticks.

3.6 Lid. A steel lid shall be furnished. It shall not be less than 0.015 inch thick, shall be provided with a handle or knob made of heat insulating material, and shall be plated to resist corrosion. Such lid shall be either portable or hinged and, when pot is not in use, shall exclude dust and dirt.

3.7 Nameplate. Each gluepot shall be equipped with an affixed nameplate marked in a legible manner with the following information:

Gluepot
Type
Size
Volts_____ Amps_____ Cycles_____
Fed. Spec. W-G-441
Manufacturer's Model No.
Manufacturer's Serial No.
NSN (To be supplied by the procuring agency.)
Manufacturer's name or trademark
Country of origin

3.8 Heating element. Each pot shall be provided with an arranged electric heating element, so located and insulated thermally as to prevent fire risk or damage to a wood support-surface when operating at the maximum temperature conditions at rated voltage. Heating elements shall be made from composition D (60 nickel, 16 chromium alloy) or composition E (80 nickel, 20 chromium alloy) conforming to ASTM B 344.

3.9 Thermostat.

3.9.1 Use. The temperature of the glue shall be automatically maintained by one or more thermostats within the working temperature limits under the conditions prescribed in 4.6.1.

3.9.2 Temperature setting. The temperature setting of the thermostat shall be the mid-point between those temperatures related to the making and breaking temperatures.

3.9.3 Working temperature limits. Working temperature limits as controlled by the thermostats shall be from 140 degrees to 200 degrees Fahrenheit (F.).

3.9.4 Sensitivity. Sensitivity of the thermostat shall be plus or minus 3 degrees F. for the temperature setting over the range of 140 degrees F. through 200 degrees F. (see 4.6.1).

3.9.5 Design. Each thermostat shall be quick break, and shall make and break contacts in the heater supply circuit at low and high predetermined temperatures of the inner pot contents. All contacts on thermostats shall be provided with hardened alloy or silver tips and arc absorbers. The mechanism on the thermostat for temperature-setting adjustment shall be both easily accessible and adjustable throughout the temperature range of 140 degrees F. to 200 degrees F..

3.10 Electrical fittings. Unless otherwise specified (see 6.1), each gluepot shall be fitted with a male, two-pole, parallel, motor-base-plug. This fitting shall be recessed to shield against drip and attached to the outer container of the gluepot.

3.11 Electrical cable. Unless otherwise specified, (see 6.1), each gluepot shall be furnished with not less than 8 feet of type SO, heavy duty, 2-wire extension cord. There shall be a shielded plug cap on one end and a cord connector body on the other end.

3.12 Insulation, heat and electrical. Heat and electrical insulation in proximity to parts subject to temperatures in excess of 49 degrees Celsius (C.) (120 degrees F.), shall be flameproof, slow-burning, heat-resisting material.

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3.13 Dielectric strength. Dielectric material shall meet the requirements of 4.6.5.

3.14 Electrical insulation resistance. The electrical insulation shall meet the resistance test of 4.6.6.

3.15 Performance.

3.15.1 Warming-up-time. The heater, when operated at rated voltage, shall distribute heat to the inner pot to melt the contents so that, when stirred, a uniform viscosity may be obtained in the time as given in 4.6.2.4.

3.15.2 Temperature differential, quarter-full pot. The heater, when operated at rated voltage, shall meet the requirements of 4.6.3.

3.15.3 Energy consumption. The heater, when operated at rated voltage, shall meet the rquirements of 4.6.4.

3.16 Castings. All iron castings shall be of uniform quality, free from blowholes, porosity, hard spots, shrinkage defects, cracks or other injurious defects. Strength and other physical properties of the castings shall be adequate throughout for the purpose intended.

3.17 Fungus resistance. When specified (see 6.1), components of the electrical equipment shall be treated to resist fungus.

3.18 Workmanship. The gluepots shall be free from cracks, rough or sharp edges, and any defects that would affect appearance, serviceability or durability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that the supplies and services conform to prescribed requirements.

4.1.1 Inspection for materials and components. In accordance with 4.1, the supplier is responsible for insuring that materials and components used were manufactured, tested and inspected in accordance with the requirements of referenced subsidiary specifications and standards to the extent specified herein, or, if none, in accordance with this specification.

4.2 Sampling for inspection and testing. Sampling shall be in accordance with MIL-STD-105. The lot size shall consist of all gluepots offered for acceptance at one time. The sample unit for end item inspection and testing shall be one gluepot. The sample unit for preparation for delivery inspection shall be one container fully prepared for delivery.

4.3 Inspection of the end item. The gluepots shall be visually inspected for design, construction, identification markings and workmanship. The inspection level shall be I with an acceptable quality level of (AQL) of 2.5 percent defective.

4.4 Testing. Each sample unit shall be tested in accordance with paragraph 4.6, as applicable. The inspection level shall be S-2 with an AQL of 2.5 percent defective.

4.5 Inspection of preparation for delivery. An inspection shall be made to determine compliance with the requirements of section 5. Defects shall be scored in accordance with table I. The inspection level shall be S-2 with an AQL of 4.0 percent defective.

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TABLE I. Classification of Preparation For Delivery Defects

Examine	Defects
Markings (exterior and interior)	Omitted; incorrect; illegible; improper size, location, sequence, or method of application.
Materials	Any component missing, or damaged.
Workmanship	Inadequate application of components such as incomplete closure of container flaps, loose strapping, inadequate stapling. Distortion of container.

4.6 Test procedures.

4.6.1 Temperature setting and sensitivity test.

4.6.1.1 Test condition. Initial temperature as per 4.6.2.1; water level as per 4.6.2.2, for type II; glue level as per 4.6.2.3; and standard thermometer (mercury-in-glass), Bureau of Standards approved.

4.6.1.2 Method of test. Heat pot for 3 hours. During the last 15 minutes immerse the thermometer approximately 3 inches and stir the glue constantly. Take temperature readings at intervals of 1 minute and plot results on graph paper, time being on the abscissa and temperature being on the ordinate. The curve must simulate a sine wave. The midpoint temperature between any peak and the adjacent valley temperature will be the temperature setting of the thermostat. The sensitivity of the thermostat shall be the total difference between the peak and valley temperatures. Adjust the temperature-setting mechanism to check the sensitivity at 140 degrees, 160 degrees, and 200 degrees F.. In no instance shall the sensitivity be more than 6 degrees F..

4.6.2 Warming-up-time test.

4.6.2.1 Initial temperature. All parts and contents shall be at room temperature, i.e., 68 degrees to 77 degrees F..

4.6.2.2 Water level. In case of type II, the water level in the jacket shall be at least up to the level of the glue or at least 1/2 inch from the top of the water compartment, whichever is the highest.

4.6.2.3 Glue level. The pot shall be filled to 1/4 inch from the brim with an 11-percent solution of grade J2 glue, conforming to MMM-A-100, which shall have been presoaked 16 hours at room temperature.

4.6.2.4 Time limits. While being stirred as necessary, the starting time of the warming-up period of the 1-quart and 2-quart sizes shall not exceed 45 minutes to attain 140 degrees F. glue temperature with the portable cover not in place. The 4-quart size shall require not more than 65 minutes for the same warming-up period with the portable lid not in place.

4.6.3 Temperature-differential test, quarter-full pot.

4.6.3.1 Repeat test of 4.6.1 except pot shall be one-quarter filled with glue.

4.6.4 Energy-consumption test.

4.6.4.1 Initial conditions. Same as for 4.6.1.

4.6.4.2 The electric energy shall be measured by a watt-hour meter of an accuracy of less than 10 percent error. The energy expenditure during 7 hours' operation, starting cold, shall not exceed that specified in table II for a 200 degrees F. temperature setting.

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TABLE II. Energy consumption

Size	Total energy expenditure	
	Type I	Type II
Quarts	Kilowatt hours	Kilowatt hours
1	0.55	0.65
2	0.80	1.10
4	1.50	2.00

4.6.5 Dielectric strength test. Each pot, while hot, shall be subjected to a dielectric test, using a 60-cycle alternating voltage of 1,000 volts, impressed between the current-carrying parts and the housing and pot, having practically a sine-wave form applied for 1 minute without rupturing the dielectric.

4.6.6 Electrical insulation resistance test. Each pot, while hot, shall be tested with a 1,000 volt megger, or equivalent instrument, where the voltage is impressed in the same manner as 4.6.5, and such electrical insulation resistance shall be at least one megohm.

5. PREPARATION FOR DELIVERY.

5.1 Preservation and packaging. Preservation, packaging, packing, and marking shall be in accordance with PPP-P-40. The level of preservation, packaging, and packing shall be A or C, as specified (see 6.1).

6. NOTES

6.1 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents.

(a) Title, date and number of this specification.
(b) Type and size required (see 1.2 and 1.3).
(c) Operating voltage required (see 3.3).
(d) Construction of inner pot if other than specified (see 3.5).
(e) Electrical fittings if other than specified (see 3.10).
(f) If electrical cable is not required other than specified (see 3.11).
(g) When fungus treatment is required (see 3.17).
(h) Level of preservation, packaging and packing (see 5.1).

MILITARY INTERESTS:

Military Coordinating Activity
Army - GL

Custodians
Army - GL
Air Force - 99

Review activity
Air Force 84

PREPARING ACTIVITY

GSA - FSS

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